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RESEARCH ARTICLE

Japanese Short Answer Grading for Japanese Language Learners Using the Contextual Representation of BERT

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ABSTRACT The automatization of grading short answers in examinations aims to help teachers grade more efficiently and fairly. The Japanese SIMPLE-O attempts to grade Japanese language learners' short answers using a dataset from a real examination. Bidirectional encoder representations from transformer (BERT), which has shown potential for natural language processing (NLP) tasks, is implemented to grade answers without fine-tuning due to the small amount of data. Two experiments are conducted in this study. The first experiment attempts to grade based on similarities, while the second classifies the answers as either correct or incorrect. Five BERT models are tested in the system, and two additional sentence BERT (SBERT) and RoBERTa models are tested for the similarity problem. The best Pearson's correlation for grading with similarities is obtained with the Tohoku BERT Base. The use of hiragana–kanji conversion improves the correlation to 0.615 for BERT and 0.593 for SBERT but does not show much improvement for RoBERTa. In the binary classification experiments, all models have an accuracy above 90%, with Tohoku BERT Large having the best performance. Even without fine-tuning, BERT can be used as an embedding method to perform binary classification with high accuracy.

INDEX TERMS Automated short answer grading, BERT, SBERT, deep learning, contextual embeddings.

I. INTRODUCTION

The short answer is one of the question types that frequently appears during examinations. It is also used in Japanese language examinations for Japanese language class students. However, the answers to these examinations are still mostly graded manually. As there are more students than teachers in the classroom setting, automatization should make the grading process more efficient and fairer. However, grading free text is more complicated than multiple choice as the reference answer, and students' answers are usually not the same. A system that can automatically grade short answers is known as the automated short answer grading (ASAG) system.

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Many studies have been conducted on the English language, while some have aimed to grade other languages [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], including the Japanese language. However, few studies have discussed ASAG for the Japanese language, especially for learners. The Japanese language is different from English, especially because it does not use the alphabet and has its own characters, such as kanji, hiragana, and katakana. Compared with English, other languages do not have as many resources, especially those that are open to the public. Thus, it may be necessary to use a personal dataset to build a certain system.

SIMPLE-O is a short-answer grading system developed by the Department of Electrical Engineering, Universitas Indonesia [13]. Initially, it was developed for Indonesian language short answers. However, since 2018, a system for Japanese language short answers has been developed,

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considering the potential and demand for Japanese studies. The grading system is mostly performed by identifying the similarities between the reference answer and the students' answer. Classifying the answers into correct and incorrect has not yet been performed.

Many methods have been studied to improve the scoring performance of the ASAG system, with or without machine learning. The method of solving the ASAG problem can be classified into regression and classification problems [14]. The regression model is graded depending on how similar the answers are to the reference, whereas classification groups the answers based on set categories.

Among these methods, the current state-of-the-art algorithm, bidirectional encoder representations from transformers (BERT) shows good potential for solving many natural language processing (NLP) problems, including ASAG. Many studies have reported that a system that implements BERT yields satisfactory results [5], [15], [16]. Some researchers have also noted that its performance is better than that of other machine learning approaches [5]. There are some state-of-the-art methods based on BERT, such as Sentence BERT (SBERT) [17], which embeds sentences instead of words, and robustly optimized BERT approach (RoBERTa) [18], which is the modification of BERT. These methods are used in certain ASAG systems [11], [19], [20], [21], [22], [23], [24].

BERT can capture the context of a text from the right and left and create richer representations [25]. When dealing with text inputs, there may be a need for tedious preprocessing when extracting the features to ensure that only important things are processed in the main algorithm. The use of a BERT-based model can reduce or even eliminate this requirement.

BERT is not restricted to the English language, as it also has multilingual models. These BERT models are trained for specific languages based on the original BERT. The Japanese language has BERT models for the language that will be used in this study. The Japanese SBERT model is trained based on this model.

This study contributes to the field of short answer grading for the Japanese language, especially for non-natives who are learning the Japanese language. It shows the characteristics of the answers of the Japanese learners who participated in the study. This study aims to develop an ASAG for the Japanese language using the Japanese SIMPLE-O dataset, which is taken from the answers of non-native Japanese learners using BERT. Various BERT-based models have been used to grade answers. Using the Japanese SIMPLE-O dataset, the system determines its potential use and how it can be used for grading based on the similarities between answers when used for classification. This study also compares the performance of Japanese BERT and its variations, SBERT and RoBERTa for the similarity scheme. The limitation of this study lies in the small amount of data in the private dataset which results in limited adjustments or fine-tuning.

This paper is organized into five sections: Introduction, Concepts, Methods, Results and Discussion, and Conclusion. The introduction presents an overview of the topic and the motivation behind this study. Section II discusses the concepts related to the study. The method and design of the system and the test scenarios are presented in Section III, and the results are discussed in Section IV. Section V presents the conclusion and future works.

II. AUTOMATED SHORT ANSWER GRADING (ASAG) CONCEPTS

This section discusses the underlying concepts of the study, including related topics such as short answer grading, embeddings, and BERT. Other related and previous studies on automated short answer grading and the Japanese SIMPLE-O itself are also examined.

A. SHORT ANSWER GRADING

Automated grading for free texts was first introduced by Page [26] for English essay examinations. Since then, many automated essay grading methods have been developed. One of them is the Japanese Essay Scoring System (JESS) [27], which aims to grade Japanese essays using latent semantic analysis (LSA). However, research on free-text answers has become increasingly specific.

Currently, there are two studies on free-text answer grading: automated essay scoring (AES) and ASAG. While AES tries to grade long essays, ASAG is used to grade short answers. According to Burrows [28], short answers usually fulfill five criteria: requiring other prior sources of knowledge, using natural language in the responses, keeping the answers relatively brief (ranging from one sentence to one paragraph), focusing on content rather than writing style for evaluation, and designing questions in a way that restricts answers to objective responses. Unlike AES, which has many holistic criteria for scoring, ASAG is usually graded by comparing students' answers with the reference answer [14].

ASAG has undergone many advancements in its algorithms. Reviews by [28] and [29] mention that the development of the ASAG system can be divided into several eras. Initially, concept mapping is used to compare students' answers with the reference key. Predictions based on similarities can be made by assessing the overlap with reference answers. In the next era, the information retrieval method focuses on feature extraction. LSA has been used in many earlier studies [8], [13], [27], [30], but this method has also been used in some recent studies [4], [6]. However, until the era of knowledge-based methods and models trained on a large corpus, no approach has considered the similarity and semantic relationships between words in sentences. Subsequently, research on ASAG begins to implement feature-based machine learning for classification and regression. Word embedding is one of the most important advancements in NLP. Machine learning methods are also greatly improved with deep learning methods,

which can capture dependencies. The current state-of-the-art is BERT, an attention-based model that can capture context. Some recent studies have used deep learning [7], [31], BERT [5], [11], [15], [16], SBERT [15], [16], [17], [19], and RoBERTa [11], [23], [24].

B. EMBEDDINGS

Embedding is a process used to represent words or sentences in numerical values or vectors [32]. Several embedding methods have been used in NLP research. A review by [33] shows that embedding methods have been growing rapidly since 2013, with the appearance of Word2Vec. Previously, researchers used one-hot encoding or Term Frequency–Inverse Document Frequency (TFIDF) to model words in the texts. However, these methods cannot capture the semantic relationships between words. Word2Vec [34] attempts to design a model that could understand the underlying semantic and syntactic similarities. It learns word vectors from a large dataset trained using a neural network. These word vectors are pre-trained and can be used in other tasks. Other similar pre-trained models are GloVe [35] and fastText [36].

Pre-trained word embeddings, such as Word2Vec and GloVe, are widely used in the study of ASAG. Studies such as [5], [12], and [37] use Word2Vec, while [20] and [38] use GloVe as the embedding method. In [39], automatic scoring using various embedding methods shows that among GloVe, Word2Vec, FastText, and Embeddings from Language Models (ELMo), Word2Vec managed to obtain the highest Pearson's correlation value. Previous studies on SIMPLE-O have used Word2Vec as the embedding method [5], [7].

However, these models may have difficulty with out-of-vocabulary words and are only able to capture the context of each word rather than the entire context [40]. With the development of deep learning, embedding methods have also been trained in deeper networks and can thus capture more context. An example of this embedding method is ELMo [40], which uses bidirectional long short-term memory (LSTM). Current state-of-the-art embeddings, such as general pre-trained model (GPT) [41] and BERT, use transformer-based architecture [42] for pre-training a large amount of data.

A scoping review of embedding methods used in the ASAG system is presented in [43]. The review groups embedding methods into four categories: word, sentence, sense, and contextual embeddings. Word embeddings represent a word based on its relationship with its surroundings [32]. Instead of word by word, a sentence can be represented by sentence embedding using methods such as Sentence-BERT [17]. Sense embedding uses word sense disambiguation techniques to capture the more precise meaning of a word as in [31]. While many ASAG studies have used word embeddings, such as Word2Vec and GloVe, more recent studies have represented the text with contextual embeddings, including ASAG, because these algorithms can capture contextual meaning [43].

C. BIDIRECTIONAL ENCODER REPRESENTATIONS FROM TRANSFORMER

BERT [25] is a language model that uses transformers based on attention mechanisms. It is pre-trained bidirectionally, unlike the left-to-right model, which can only see the previous token. This model aims to build general language representations that can minimize the need for a task-specific architecture. There are two frameworks for BERT, namely pre-training and fine-tuning, which use the same architecture.

The pre-trained BERT trains the model using unlabeled data. This is performed with a masked language model, which masks several words in the sequence and predicts the words bidirectionally, and next-sentence prediction, which is a binary classification that predicts whether the sentence is the next sentence or not. In fine-tuning, the parameters from pre-training can be fine-tuned using labeled data from downstream tasks. One of the benefits of BERT is its ability to conduct transfer learning. Through pre-training, it can learn vocabulary, grammar, and other linguistic patterns.

The input for BERT is represented by tokenizing the words. Then, a special token is added. These special tokens are [CLS], which marks the beginning of a sequence or sentence, and [SEP], which is placed at the end of a sentence and can also function as a separator between sentences. If the inputs are padded, a special token is used [PAD]. Subsequently, the inputs are changed to their token IDs and processed using the BERT encoder. The results are token embeddings with 768 hidden sizes. This output can be used for other purposes, such as classification. The process is illustrated in Fig. 1.

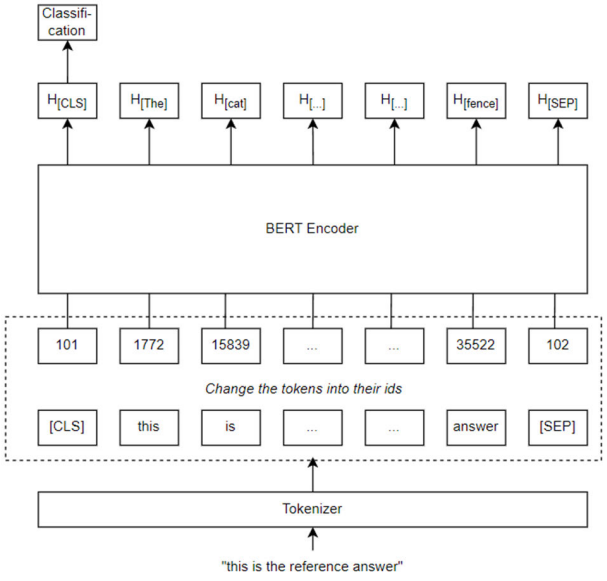


FIGURE 1. BERT input representation example for a single sentence [25].

D. RoBERTa

RoBERTa is a transformer-based embedding model by Facebook AI which optimizes the existing BERT model [18]. It improves the training process by making some modifications.

With a larger dataset, the training is done longer and with bigger batches. While pre-training in BERT consists of a masked language model and next-sentence prediction, RoBERTa removes the next-sentence prediction. The masking process uses dynamic masking instead of the original single static masking. RoBERTa also differs from BERT in terms of learning sequence as RoBERTa uses full-length sequences for pre-training.

E. SENTENCE BERT

SBERT was introduced in 2019 as an embedding method for sentences based on a modified BERT [17]. The model is built with a Siamese BERT architecture. Fig. 2 illustrates the SBERT architecture used to determine the similarities between two sentences. The embeddings from each BERT model are pooled using mean pooling as the default setting. The sentences generate sentence embeddings u and v, which can be processed further using cosine similarity.

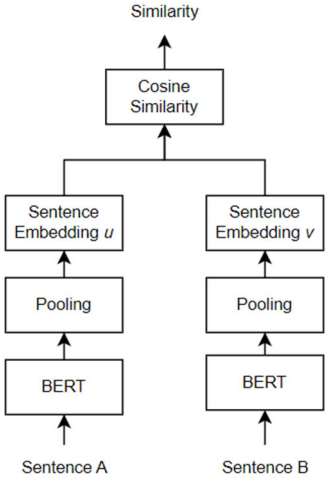


FIGURE 2. SBERT architecture for similarity problems [17].

F. RELATED STUDIES

Many studies have been conducted to develop an ASAG model, with some implementing BERT in their systems. Zhu et al. [15] use a pre-trained fine-tuned BERT to encode sentence pairs consisting of reference and student answers. The proposed system adds a semantic refinement layer with bidirectional LSTM (BiLSTM) and a capsule network, as well as a semantic fusion layer, before predicting the score. According to the results, the BERT model can achieve good performance through fine-tuning, which allows the LSTM and capsule networks to gain a better understanding of the words. Garg et al. [16] train the tokens from the reference answer and students' answer pair using the BERT model to predict the score. The authors also attempt to determine the similarities between the answers using BERT and cosine similarity. The predicted grades and similarities are integrated using an artificial neural network to obtain the final grade. The model reaches a Pearson's correlation value of 0.777 in the Mohler dataset.

Some studies have attempted to categorize student answers into classes. Wijaya [10] classifies Indonesian short answers using BERT. The questions and answers are concatenated and processed using pre-trained BERT. With fine-tuning, the model can predict labels with 0.94 precision, 0.96 recall, and 0.95 F1-score. Alhamed et al. [44] propose a system that incorporates BERT with BiLSTM to grade the examinations of information systems. The dataset consists of three labels: 0, 0.5, and 1. The system achieves an accuracy of 85% and a macro average for precision and recall of 82%.

The superiority of BERT over other methods has been demonstrated in many studies. In [9], Schleifer et al. attempt to grade short answers in the field of biology in the Hebrew language. This study compares the performance of the BERT classifier with a convolutional neural network (CNN)-based system. The results show that the BERT-based model outperforms CNN, which uses Word2Vec as an embedding method. Salim et al. [5] compare a fine-tuned BERT-based model with ridge regression to grade short answers in the Indonesian language. In this study, the system with fine-tuned BERT manages to outperform ridge regression with ALBERT, with a Pearson's correlation value of 0.95, while ridge regression has a value of 0.92.

In addition to BERT, its alternatives such as RoBERTa and SBERT have also been applied in recent research on ASAG. A study attempts to use ASAG for grading clinical examinations using SBERT and contrastive learning based on the similarities between the embeddings of the answers [19]. In [20], the authors compare instance-based methods using logistic regression (LR) and BERT with similarity-based method using SBERT. In this case, BERT and SBERT outperform LR. Another comparison study is conducted in [21], in which the authors compare the SBERT model with the BERT and GloVe models in the Mohler dataset. Although it is better than GloVe, SBERT's performance is still lower than that of BERT. SBERT has also been used in languages other than English. The study in [22] uses SBERT for the ASAG system in the Indonesian language. Different results are shown in a study using an Indian dataset [11]. The study compares many transformer-based models, such as BERT, GPT-2, XLNet, SBERT, and RoBERTa for features extraction. The study shows that RoBERTa and SBERT outperform BERT in terms of RMSE and correlation. Some studies highlight the implementation of RoBERTa in ASAG. In [23], Siamese RoBERTa bi-encoder and cosine similarity to calculate the similarity scores between student answers and reference answers and then classify those as either correct or incorrect. The method is able to reach 0.69 for accuracy and 0.7 for F1 score. The study of the effectiveness of RoBERTa is also done in [24] which uses RoBERTa Large and RoBERTa Large with fine-tuned multi-genre natural language interference (MNLI).

Attempts to grade Japanese free-text answers have been mentioned in some studies, beginning with JESS [27]. Similar research is conducted by Hirao et al. [45] on grading free-text answers in the Japanese language for non-native learners.

TABLE 1. Example of the japanese SIMPLE-O dataset.

Question	2020年東京オリンピックのマスコットはどうやって決められましたか。 (How was the mascot for the Tokyo Olympics 2020 decided?)
Reference Answer	全国の小学生の投票で決められました。 (It was decided through voting by elementary schoolers across the country.)
Students' Answers	小学校のとうひょうで決められました。 (It was decided through voting by elementary schools.) ぜんこくの小学生のとうひょうで決まります。 (It is decided through voting by elementary schoolers across the country.)

The input is a dataset of essays written by overseas Japanese learners. According to the study, BERT is more robust and achieves the highest score compared with linear SVR, linear regression, random forest, and LSTM. Another study on ASAG for the Japanese language is performed in [46]. The researchers use a neural baseline with BiLSTM and attention mechanism and another baseline using SVR. The dataset in this study uses rubrics based on both analytic and holistic scoring criteria.

SIMPLE-O has been developed for the Japanese language since 2018. Previous studies have used various methods, such as LSA [8], Winnowing [8], CNN-LSTM [1], and CNN-BiLSTM [7]. The LSA and Winnowing methods compare the similarities between the reference answer and students' answers using cosine similarity. CNN-LSTM uses fastText as the word embedding method, while CNN-BiLSTM uses Word2Vec. The predictions are trained based on the similarities from the Manhattan distance.

III. METHODS

This section discusses the method used in the study, including the dataset and the designed system. Two scenarios are experimented with: scoring by measuring the similarities between the reference answer and students' answers and scoring by classifying whether the answer is correct or incorrect. The results are evaluated to determine the system's performance.

A. DATASET

The system inputs use the SIMPLE-O dataset obtained from an actual mid-term examination. The students participating in this examination are Indonesian students who enrolled in the intermediate Japanese class. This dataset is a private dataset from the lecturer of the class and is not published anywhere.

The dataset consists of five questions that ask about the definition of a term. Each question has a different number of reference answers. Questions numbers 1 and 3 have two reference answers, question numbers 2 and 4 have three reference answers, and question number 5 has seven reference answers. Forty-three students participated in this examination. In total, the dataset contains 215 answers. Examples of the questions, the reference answers, and the students' answers are presented in Table 1.

The answers, both the reference and the students' answers, are written in Japanese. As the examination is for intermediate-level Japanese learners, katakana, hiragana, and kanji are allowed. SIMPLE-O is intended for Japanese language learners. Thus, it is possible for students to answer in the hiragana form of a kanji word. This nature may be slightly different from sentences written by native Japanese speakers. The answers are also manually graded by experts, in this case, the class lecturer. The questions in the dataset are originally part of the same test with a maximum score of 100, with a proportion of the scores divided among the five questions. The maximum possible score for each question is 20. In this study, each question is treated separately. Thus, the score range for each question is normalized to 0–100. This is also done for easier analysis and readability as it allows the data to be in a similar range with the outputs of the sentence similarity experiment and the exam's score range.

For the classification, label 1 is for correct answers at a range of 90–100, while label 0 is for answers scoring below this range. The scores in label 1 are grouped as correct because the students answered closely to the answer that the teacher considered correct and only made slight to no mistakes in grammar. In total, there are 731 pairs of reference answers and students' answers to classify, with 345 answers labeled 1 or correct and 386 answers labeled 0 or incorrect.

B. PREPROCESSING

Before the dataset is represented by the embedding method, each sentence undergoes preprocessing. In the preprocessing step, the texts are cleaned of punctuation marks, such as ○,-, 「」, and whitespaces. Romaji or the alphabet are also removed, if there are any. This is performed in case there are students who answer in either Romaji or other languages, such as the students' first language, Indonesian, which uses the alphabet.

C. PRE-TRAINED MODELS

After preprocessing, the data are tokenized and represented using pre-trained BERT. As the dataset is too small for fine-tuning, the experiment does not include it in the process. Five pre-trained BERT models are tested with the SIMPLE-O dataset. In addition, three RoBERTa and two SBERT models are tested for comparison with the BERT models. However, these models are not used for the classification experiment. The models used in this study are presented in Table 2. Japanese BERT is trained with Japanese Wikipedia. The architecture follows the original BERT. The base models and the multilingual model have 12 layers, 768 dimensions of hidden states, and 12 attention heads, while the large model has 24 layers, 1,024 dimensions of hidden states, and 24 attention heads.

TABLE 2. BERT Pre-trained models in this study.

Model	Abbreviation
cl-tohoku/bert-base-japanese-v3[1]	Tohoku BERT Base
cl-tohoku/bert-base-japanese-char-v3[2]	Tohoku BERT Base (char)
cl-tohoku/bert-large-japanese-v2[3]	Tohoku BERT Large
cl-tohoku/bert-large-japanese-char-v2[4]	Tohoku BERT Large (char)
bert-base-multilingual-cased[5]	BERT Base Multilingual
sonoisa/sentence-bert-base-ja-mean-tokens-v2[6]	SBERT Sonoisa
sentence-transformers/distiluse-base-multilingual-cased-v2[7]	SBERT Multilingual
rinna/japanese-roberta-base[8]	Rinna RoBERTa Base
nlp-waseda/roberta-base-japanese[9]	Waseda RoBERTa Base
nlp-waseda/roberta-large-japanese[10]	Waseda RoBERTa Large

D. SENTENCE SIMILARITY EXPERIMENT IN SIMPLE-O DATASET

This scenario determines the performance of the system when used for scoring based on the similarities between two sentences. In the two sentences, the students' answers and the reference answers undergo preprocessing and are then processed using BERT. The outputs are the embedding for each sentence. There are many ways to obtain a representation of a sentence from BERT. This study takes the representation from the CLS and compares it to the results when using the mean from the last hidden state. The similarities of the outputs are measured with cosine similarity. The output of the system is multiplied by 100 to get the range of 0-100. This value becomes the system-generated score for the students' answers. This process is shown in Fig. 3. For comparison, the system also experiments with the SBERT model using the same scheme. SBERT is pooled using the default pooling method, which uses the average value of the word embeddings.

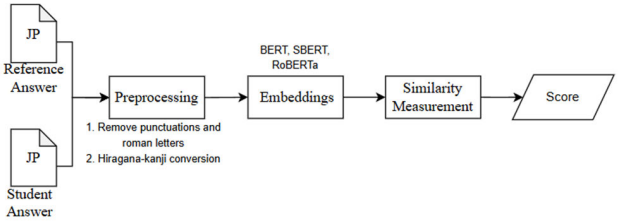


FIGURE 3. General flow design of the similarity experiment.

[1]https://huggingface.co/cl-tohoku/bert-base-japanese-v3
[2]https://huggingface.co/cl-tohoku/bert-base-japanese-char-v3
[3]https://huggingface.co/tohoku-nlp/bert-large-japanese-v2
[4]https://huggingface.co/cl-tohoku/bert-large-japanese-char-v2
[5]https://huggingface.co/bert-base-multilingual-cased
[6]https://huggingface.co/sonoisa/sentence-bert-base-ja-mean-tokens-v2
[7]https://huggingface.co/sentence-transformers/distiluse-basemultilingual- cased-v2
[8]https://huggingface.co/rinna/japanese-roberta-base
[9]https://huggingface.co/nlp-waseda/roberta-base-japanese
[10]https://huggingface.co/nlp-waseda/roberta-large-japanese

An evaluation of the system scores is required to determine the system's performance. In this similarity experiment, the evaluation is conducted by calculating the correlation value between the scores. Using the manual score provided in the dataset, the correlation between the scores and the system's scores can be calculated using Pearson's correlation.

E. CLASSIFICATION EXPERIMENT IN THE SIMPLE-O DATASET

The second scenario is a binary classification problem. Instead of using the similarities as a score, this experiment attempts to determine the performance of BERT when used for predicting correct or incorrect answers using the SIMPLE-O dataset. After preprocessing, the reference answer and students' answers are combined to obtain the contexts of both sentences and then processed with BERT. Only BERT models are tested in this experiment. Using the classification output, the model performs a binary classification in the combined sentence. The flow of the system used in this experiment is shown in Fig. 4.

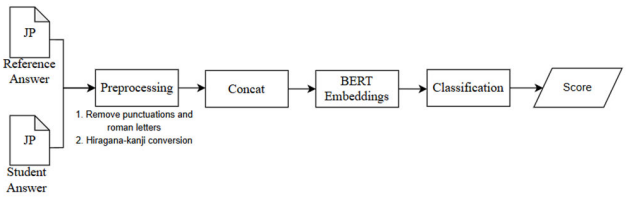


FIGURE 4. General flow design of the classification experiment.

During the training for the prediction, the dataset is split into data training and testing, with a proportion of 80:20 due to the small amount of data. Hyperparameters are important things to consider for the system to be able to output predictions as intended. Various settings, such as the learning rate, batch size, warm-ups, and the number of epochs were tested in a preliminary experiment. In this research, the training process has hyperparameters of 5 epochs, a learning rate of 2e-5, a batch size of 8, and a warm-up of 0.1. Considering the relatively small number of data, a small epoch and a small batch size are sufficient to train the model without resulting in overfitting.

The evaluation of the system's score uses various metrics, such as accuracy, precision, recall, and F1 score. These metrics can be calculated based on the true negatives (TN), true positives (TP), false positives (FP), and false negatives (FN) of the predictions, as illustrated in Fig. 5. The formula for each metric is expressed in Equations (1)–(4).

$$\text{Accuracy} = \frac{\text{TP}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{1}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{2}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{3}$$

$$\text{F1} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$

TABLE 3. The questions in the dataset.

Number	Question
1	2020年東京オリンピックのマスコットはどうやって決められましたか。 (How was the mascot for the Tokyo Olympics 2020 decided?)
2	ハラルというのは何ですか。 (What is 'halal'?)
3	生前退位というのは何ですか。 (What is abdication?)
4	衆院選というのは何ですか。 (What is the House of Representation election?)
5	Uターン就職というのは何ですか。 (What is a U-turn job?)

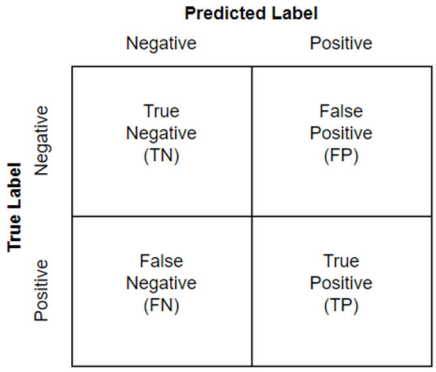


FIGURE 5. Confusion matrix.

IV. RESULTS AND DISCUSSION

This section reports the results of the experiments conducted based on the designed system. The system performance determines whether BERT is suitable for the current dataset and which parts need improvement. However, the dataset itself can affect performance. Thus, this section also discusses the characteristics of the dataset from the Japanese SIMPLE-O.

A. DATASET ANALYSIS

Five questions in the dataset ask about a general understanding of a term. These questions are presented in Table 3. The respondents belong to a class consisting of 43 students.

Looking at the students' answers, some students did not answer some questions; this automatically obtains zero points. As the dataset is obtained from an actual examination, the score distribution for each question is imbalanced.

Fig. 6 shows the distribution of the score for question number 1. There are two reference answers to this question. It can be inferred from the graph that the students' answers for number 1 lean toward the correct answers. Although there are no empty answers, there are many variations in the scores. Three students score 25 points, which is far lower than the other scores.

A different case is shown in Fig. 7 for question number 2. There are three reference answers to this question. One

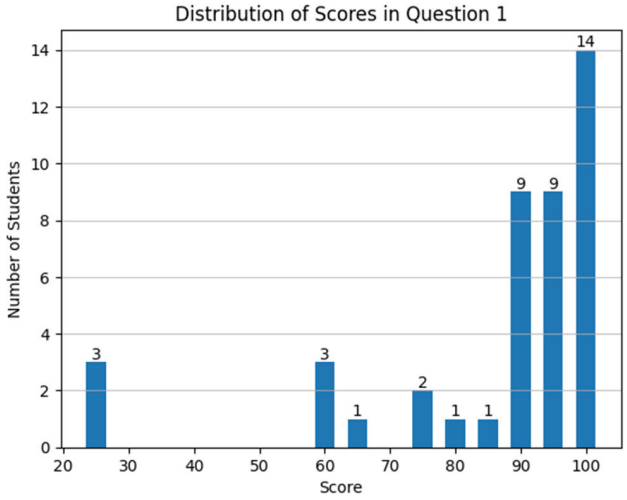


FIGURE 6. Score distribution for question number 1.

student who received a score of 0 left the answer empty. Unlike question number 1, there are more answers with scores below 90 for this question. It also has more variation in the scores, and many students obtained a score of 25.

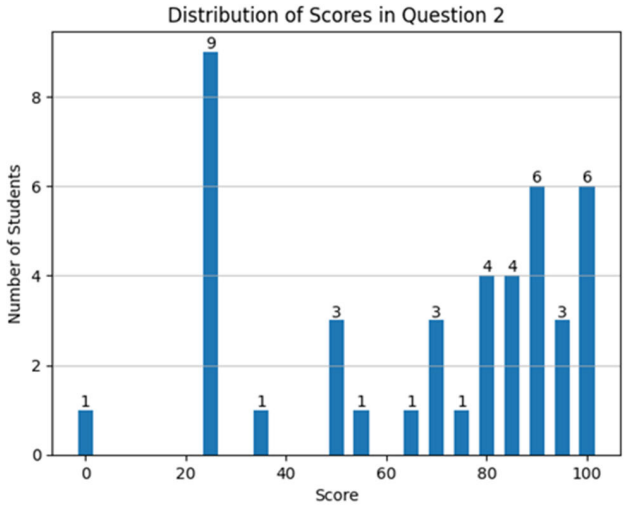


FIGURE 7. Score distribution for question number 2.

Fig. 8 shows the graph for the answers to question number 3. Similar to question number 1, the answers for question number 3 are also mostly above 90. There are two reference answers to this question. Three students left empty answers. Many students managed to get a full score on this question.

The answers to question numbers 4 and 5 are similar in their distribution. Question number 4 has two reference answers, and question number 5 has seven. Both questions have more incorrect answers than correct ones. As shown in Figs. 9 and 10, there are many students with a score of 25.

The data show that there is always a score of 25 for every question. This score is for students who try to answer, even

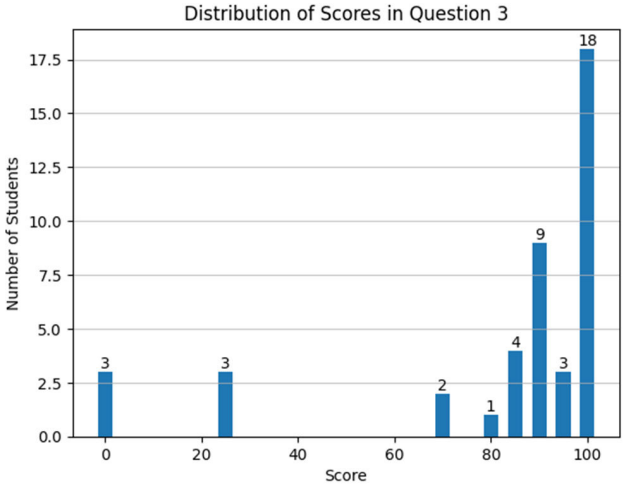


FIGURE 8. Score distribution for question number 3.

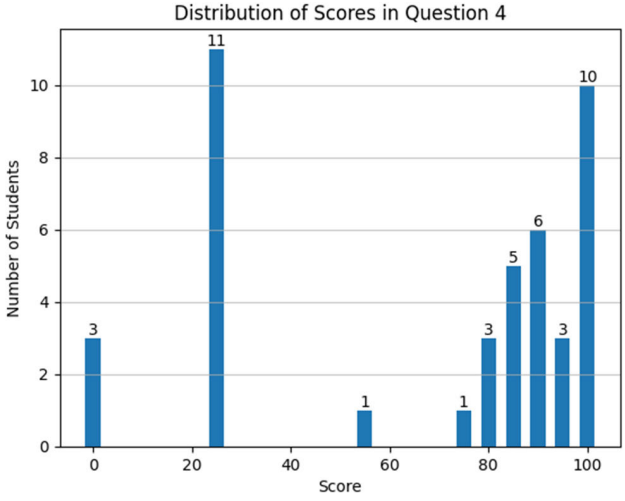


FIGURE 9. Score distribution for question number 4.

though the answer is far from the reference answer. When the answers are manually graded, the lecturer or teacher may still give a score, similar to what happens here when students do not completely leave the answer blank to appreciate students' effort to write. However, repeating the question or answering it with an unrelated answer still counts as not answering.

B. SIMILARITY EXPERIMENT RESULTS

Following the system design and scenario, experiments on Japanese SIMPLE-O are conducted using various BERT pre-trained models. In the first experiment, students' answers are compared with the reference answers. Table 4 shows Pearson's correlation for each question with the five BERT models, three RoBERTa modes, two SBERT models, and the average correlation for each model.

Based on the results, Tohoku BERT Base has the best average correlation, followed by Waseda RoBERTa Large

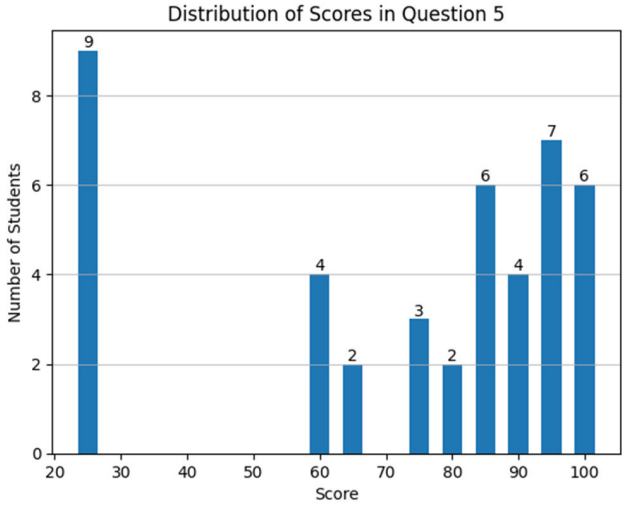


FIGURE 10. Score distribution for question number 5.

and Tohoku BERT Large. The character-based models tend to exhibit a lower correlation than the regular model, even the multilingual model. Using the mean of the embeddings instead of the CLS shows better results in the case of the SIMPLE-O Japanese system.

In this case, the Tohoku BERT Base (char) and Tohoku BERT Large (char) perform below the regular model. The same is true for the BERT Base Multilingual. The tokenization process may be one of the factors that contribute to the results. Tohoku BERT Base uses MeCab to split the text, followed by tokenization into subwords. The char models also use MeCab to divide the text but tokenize it into character levels. An example of the answer to question number 5 is shown in Fig. 11. Through this example, it is evident that BERT Multilingual suffers from tokenizing the input, which is mainly written in hiragana, and decides to take the whole answer instead.

Student Answer:
"そつぎょうした人がそつぎょうあとでじもとでしごとすることです。"
(*sotsugyoushitahitogasotsugyouatodejimotodeshigotosurukotodesu*).
(A graduate works in his hometown after graduation.)

Tohoku BERT Base and BERT Large:
"[CLS]", "そつぎょう", "し", "た", "人", "が", "そつぎょう", "あと", "で", "じ", "もと", "で", "しごと", "する", "こと", "です", "[SEP]"

Tohoku BERT Base Char and BERT Large Char:
"[CLS]", "そ", "つ", "ぎ", "ょ", "う", "し", "た", "人", "が", "そ", "つ", "ぎ", "ょ", "う", "あ", "と", "で", "じ", "も", "と", "で", "し", "ご", "と", "す", "る", "こ", "と", "で", "す", "[SEP]"

BERT Multilingual:
"[CLS]", "そつぎょうした人がそつぎょうあとでじもとでしごとすることです", "[SEP]"

FIGURE 11. Example of SIMPLE-O BERT input tokenization.

In the sentences that contain kanji, BERT Multilingual tokenizes kanji as a character. For example, the kanji 学生 (gakusei), which means a student, is tokenized as "学生" with Japanese BERT Base and Large but is tokenized as "学",

TABLE 4. BERT, RoBERTa, and SBERT results of the similarity experiment.

Model	Question Number										Average	
	1		2		3		4		5			
	CLS	Mean	CLS	Mean	CLS	Mean	CLS	Mean	CLS	Mean	CLS	Mean
Tohoku Bert Base	0.564	**0.67**	0.286	0.406	0.841	0.853	0.546	0.548	**0.35**	0.265	0.517	**0.548**
Tohoku Bert Base (char)	0.471	0.662	0.376	0.427	0.864	0.864	0.51	0.513	0.307	0.087	0.506	0.511
Tohoku Bert Large	0.518	0.492	0.405	0.449	0.846	0.856	0.567	0.553	0.22	0.242	0.511	0.518
Tohoku Bert Large (char)	0.669	0.586	0.405	0.396	0.86	**0.87**	**0.586**	0.563	0.056	0.063	0.515	0.496
BERT Base Multilingual	0.438	0.586	0.202	**0.46**	0.86	0.866	0.508	0.4	0.188	0.222	0.439	0.507
Rinna RoBERTa Base	0.434		0.372		0.86		0.473		0.288		0.485	
Waseda RoBERTa Base	0.344		0.164		0.663		0.492		0.392		0422	
Waseda RoBERTa Large	0.521		0.392		0.837		0.524		0.341		0.525	
SBERT Sonoisa	0.614		0.414		0.696		0.36		0.3		0.477	
SBERT Multilingual	0.64		0.278		0.666		0.312		0.253		0.43	

TABLE 5. The effect of the use of hiragana on similarity.

Sentence	Similarity (%)		
	SBERT	RoBERTa	BERT
全国の小学生の投票で決められました	100	100	100
ぜんこくの小学生の投票で決められました	91.957	98.7	98.856
ぜんこくのしょうがくせいの投票で決められました	59.733	96.88	95.122
ぜんこくのしょうがくせいのとうひょうで決められました	36.512	95.476	92.196
ぜんこくのしょうがくせいのとうひょうできめられました	23.674	94.343	88.19

"生" with BERT Multilingual, similar to Japanese BERT Base Char and Large Char.

Among the RoBERTa models, Waseda RoBERTa Large has considerably higher average correlation than the base models. Meanwhile, its base model has the lowest correlation of all tested pre-trained models with a value of 0.422. The system is also tested with SBERT. The Japanese SBERT Sonoisa and SBERT Distiluse Multilingual models are used in the experiment. SBERT Sonoisa generally performs better than SBERT Multilingual. However, the correlation of the system with SBERT is below most of the BERT models. SBERT Sonoisa achieves a Pearson's correlation value of 0.477, while SBERT Multilingual only has a value of 0.43, which is the lowest correlation after Waseda RoBERTa Base.

C. HIRAGANA–KANJI CONVERSION

One of the reasons for the lack of performance is that the dataset allows for any form of Japanese characters. However, the hiragana form of a kanji word may be recognized as a different word from kanji. Moreover, some kanjis have the same hiragana.

An example of how the use of hiragana can affect similarity is shown in Table 5. An example is taken from the reference answer written in kanji. From this, kanji words are changed into their hiragana forms. The underlined words are the kanji words that have been changed to hiragana. The changed sentences are compared with the original reference answer. The models used are SBERT Sonoisa, Waseda RoBERTa Large, and Tohoku BERT Base. It can be observed that similarity decreases with the number of changed words. This is particularly evident in the SBERT results. BERT and RoBERTa also show lower similarity scores, but the numbers are still considerably high, especially for RoBERTa. However, the same experiment is done with Waseda RoBERTa Base model, the similarity scores are 100, 93.128, 77.335, 65.298, and 63.168 which show a significant decrease from the change. In SBERT's case, the sentence with all hiragana only has a similarity of 23.674 out of 100 with the original sentence, even though the two sentences are the same with different forms.

To solve this problem, a hiragana to kanji conversion is implemented. The hiragana words in the students' answers are converted into respected kanji words. This process is performed during preprocessing before being represented by the embedding method. Table 6 shows Pearson's correlation values for BERT, RoBERTa, and SBERT with the hiragana–kanji conversion. The results for both BERT and SBERT show an improvement in Pearson's correlation while RoBERTa is relatively unaffected by the changes. For RoBERTa, using base or large is more important in improving the performance than the hiragana-kanji conversion. However, the result is still below BERT and SBERT even when conversion is applied.

In the case of BERT, the use of conversion considerably increases the correlation between the system scores and the manual scores. The model with the best performance is the Tohoku BERT Base, with its average Pearson's correlation increasing from 0.548 to 0.615. In the case of SBERT, the addition of the hiragana–kanji conversion generally improves the correlation value. This is especially evident in question numbers 3 and 4. This is most likely due to many students writing the hiragana form instead of the kanji form. SBERT Sonoisa obtains an average correlation of 0.593, which is near BERT's 0.615 correlation.

TABLE 6. BERT, RoBERTa, and SBERT Results of the Similarity Experiment with the Hiragana–Kanji Conversion.

Model	Question Number										Average	
	1		2		3		4		5			
	CLS	Mean	CLS	Mean	CLS	Mean	CLS	Mean	CLS	Mean	CLS	Mean
Tohoku Bert Base	0.615	**0.796**	0.358	0.391	0.868	0.865	0.563	0.565	0.345	**0.456**	0.550	**0.615**
Tohoku Bert Base (char)	0.489	0.764	0.409	**0.529**	0.862	0.864	0.531	0.565	0.278	0.258	0.514	0.596
Tohoku Bert Large	0.572	0.63	0.464	0.466	0.851	0.865	0.572	0.557	0.372	0.426	0.566	0.589
Tohoku Bert Large (char)	0.643	0.737	0.458	0.414	0.86	0.88	**0.599**	0.605	0.23	0.258	0.558	0.579
BERT Base Multilingual	0.611	0.677	0.156	0.426	0.863	0.906	0.539	0.56	0.267	0.289	0.487	0.572
Rinna RoBERTa Base	0.596		0.188		**0**.891		0.585		0.266		0.505	
Waseda RoBERTa Base	0.397		0.185		0.646		0.419		0.348		0.399	
Waseda RoBERTa Large	0.592		0.348		0.832		0.526		0.442		0.548	
SBERT Sonoisa	0.705		0.402		**0.91**		0.587		0.359		0.5926	
SBERT Multilingual	0.675		0.346		0.801		**0.599**		0.291		0.5424	

Correlation-wise, BERT may perform better than SBERT and RoBERTa. However, this may be due to the data being biased toward high-scoring answers. In general, the system with BERT shows a good correlation when many students answer correctly or have a score above 90.

Some good correlations are shown in question numbers 1, 3, and 4. Many students received the maximum score for these questions. All models obtain a strong correlation for question number 3 among all the questions. The correlations for question numbers 2 and 5 are weak, especially question number 5. Fig. 12 shows the comparison between the manual score and the system's score for question number 3, while Fig. 13 shows the comparison of scores for question number 5.

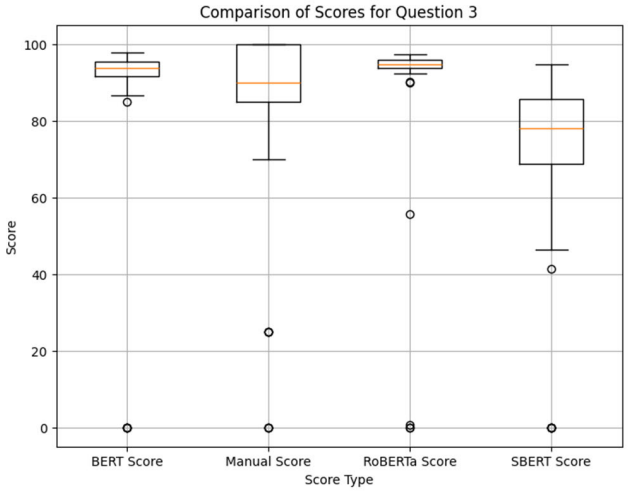


FIGURE 12. Comparison between the manual and the system scores for Question number 3.

According to the figures, the outputs of the cosine similarity on the BERT's system are distributed at the score range of 80–100, resulting in a higher median than the manual score. A similar thing also happened in RoBERTa's scores. The system also gives high scores for the answers manually marked as low scores. This not only happens in the answers to question numbers 3 and 5 but also to other questions. Currently, the system is unable to provide lower marks. This

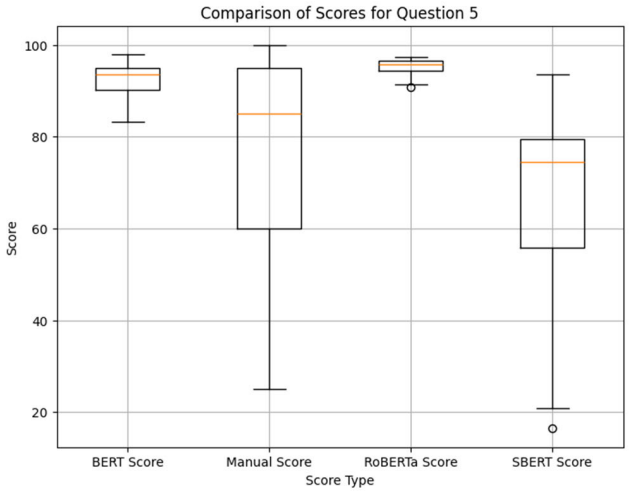


FIGURE 13. Comparison between the manual and system scores for question number 5.

may have occurred because of the similarity measurement. As the sentences also contain particles, they may be counted as similarities using cosine similarity. Therefore, it is difficult to obtain a lower score. However, obtaining full marks will only be applied to students' answers that are the same as the reference answer.

While BERT and RoBERTa tends to give higher scores, SBERT seems to struggle with giving the correct answer a perfect score. Unlike BERT and RoBERTa, SBERT can score answers with lower scores. However, as shown in Figs. 12 and 13, SBERT has a lower median than the manual score, and the distribution of scores can be even lower than what it is supposed to give. In this case, the use of different words in the compared sentences has a greater effect on the SBERT system than on the BERT system.

The performances of the BERT, RoBERTa, and SBERT systems can be determined from the standard deviation of the scores, as shown in Table 7. The data are taken from the best-performing BERT, RoBERTa, and SBERT models in this study. The low standard deviations of the system scores in the BERT and RoBERTa systems show that the scores

TABLE 7. Standard deviation for each question.

Question number	BERT	RoBERTa	SBERT	Manual
1	4.090	2.4	23.414	20.48
2	3.52	2.437	21.678	29.487
3	27.605	28.035	25.672	29.746
4	22.74	24.216	24.242	35.056
5	3.202	1.696	18.214	27.35

TABLE 8. Results of the classification experiment.

Model	Accuracy	Precision		Recall		F1	
		Incorrect	Correct	Incorrect	Correct	Incorrect	Correct
Baseline	93..88	0.97	0.90	0.83	0.95	0.95	0.92
Tohoku Bert Base	98.64	1	0.97	0.98	1	0.98	0.99
Tohoku Bert Base (char)	97.28	1	0.94	0.96	1	0.98	0.97
Tohoku Bert Large	**100**	**1**	**1**	**1**	**1**	**1**	**1**
Tohoku Bert Large (char)	99.32	1	0.98	0.99	1	0.99	0.99
BERT Base Multilingual	93.88	1	0.87	0.9	1	0.95	0.93

generated by the system tend to have fewer variations and are centered around the mean value. These values are generally far smaller than the manual and SBERT scores. On the other hand, SBERT is closer to the manual standard deviation. This means that SBERT can score with a wider range of scores than BERT-based or RoBERTa-based systems.

D. RESULTS OF THE CLASSIFICATION EXPERIMENT

The second experiment is conducted to classify the results into correct and incorrect answers. The previous experiment demonstrates a good correlation between the system and manual scores. However, the scoring range is very large, and direct scoring with similarity measurements shows that the system has difficulty in scoring lower scores or full marks. The alternative for the scoring method is binary classification.

The data are split into data training and data testing at a ratio of 80:20, resulting in 584 data trains and 147 data tests. The system performance in classifying SIMPLE-O data is measured in terms of accuracy, precision, recall, and F1-score. As the baseline, a simple classification method with TFIDF and Random Forest is used. The results of the BERT-based methods and their comparison to the baseline are presented in Table 8.

The five BERT models exhibit very good performance, with accuracies close to 100 and above the baseline. Tohoku BERT Large obtains a value of 1 in the precision, recall, and F1-scores. In this case, large models generally perform better than smaller models. BERT Large is pre-trained on a larger parameter so that it can capture more patterns. However, this performance is achieved in exchange for more memory and processing time. Although the metrics show promising results, the data are far too small and lack variation.

TABLE 9. The effect of learning rate on the model.

Learning Rate	Precision	Recall	Accuracy	F1-Score
2e-8	0.40	0.40	0.44	0.40
2e-5	0.89	0.91	0.90	0.87
2e-3	0.30	0.50	0.61	0.38
2e-1	0.30	0.50	0.61	0.38

The parameters for the experiment in Table 7 are 5 epochs, a learning rate of 2e-5, and a warm-up of 0.1. Some other parameters were also tested, such as the number of epochs, learning rate, batch size, and warm-up. Considering the small data, epochs with the values of 1, 2, 3, 4, and 5 are tested. As seen in Fig. 14, the epochs of 4 and 5 already show a small loss which is enough for this experiment.

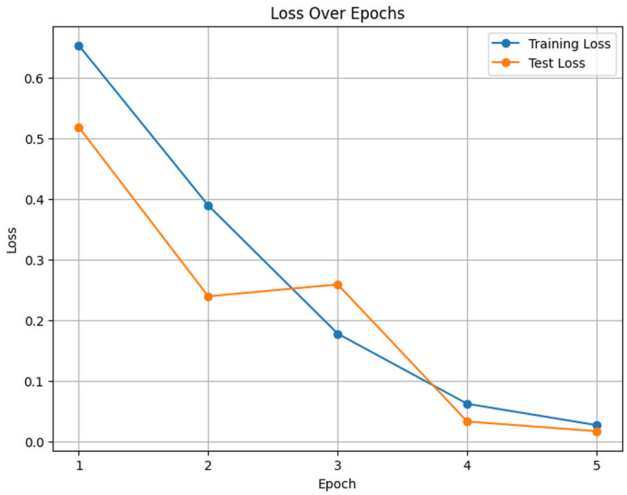


FIGURE 14. Loss over epochs graph.

The results of the experiment on learning rate are shown in Table 9. In this experiment, the epoch of 3 is used to save resources. From the table, it can be seen that the learning rate 2e-5 managed to get the best results. The learning rate of 2e-1 is too fast for the model to learn. Meanwhile, 2e-8 is too slow for the model to learn which results in slow processing time, high loss, and low accuracy.

The experiment also attempts to find the effect of the batch size. Batch sizes of 8, 16, and 32 are experimented with using a 2e-5 learning rate. The results show that a batch size of 8 performs better than a batch size of 16 and 32. The system's performance degrades with the higher batch size. The system with a batch size of 8 managed to reach 0.9 accuracy. Meanwhile, a batch size of 16 has an accuracy of 0.84 while a batch size of 32 only managed to get 0.78 accuracy. Other than the epoch, learning rate, and batch size, the performance of the system with various warm-ups is also tested. Warm-up is usually used to prevent early overfitting. However, the

experiment shows that warm-up does not have much impact on the model as there is only a small to no change in accuracy and F1-score.

V. CONCLUSION

An experiment on SIMPLE-O with several pre-trained BERT models for the Japanese language shows that BERT has a good correlation with manual scores in the ASAG system. For similarity problems, BERT, RoBERTa, and SBERT show good potential for use in this system. Among the pre-trained BERT, RoBERTa, and SBERT systems in the experiments, Tohoku BERT Base obtains the best results, with a Pearson's correlation of 0.548. The addition of the hiragana–kanji conversion resulted in a correlation of 0.615 for the Tohoku BERT Base and 0.593 for SBERT but does not have a significant change for RoBERTa. The increased performance shows that how the words are written, whether in hiragana or kanji, can affect the performance of the system. In the case of RoBERTa, the large model is better than the base model and it is able to be implemented without hiragana-kanji conversion. The use of BERT for classification also shows promising results, with the highest training accuracy of 99.83%, test accuracy of 100%, and 1 for precision, recall, and F1-score. Through these experiments, some possible ways to score students' answers can be obtained: by finding their similarities with the reference answer or by classifying them.

Although the results are good, it should be noted that the dataset used was small and biased. More data and variations are required for more extensive research to address more problems. It may be a good idea to reduce the range of scores so that the data are not spread out too much. Regarding the data in the Japanese SIMPLE-O dataset, some aspects need attention. The data also include wrong answers with 'pity scores', which require special treatment during grading. In addition, the answers written in all hiragana characters can affect the performance of the system to some degree.

The Japanese SIMPLE-O system is still being improved to be more useful and precise in grading Japanese answers in examinations. BERT, RoBERTa, and SBERT show good potential for use in this system. While paying attention to context, the system also needs to identify mistakes in grammar and writing style to meet the needs of grading Japanese language learners' short answers. In the future, incorporating the state-of-the-art language model with other NLP methods to grade the system may improve system performance.

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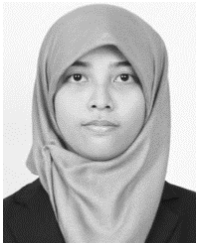
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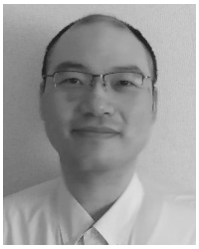


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